

HR EMPLOYEE ATTRITION PREDICTION & RETENTION STRATEGY

A MACHINE LEARNING APPROACH TO IDENTIFYING AND RETAINING AT-RISK TALENT

SDA Bocconi
School of Management

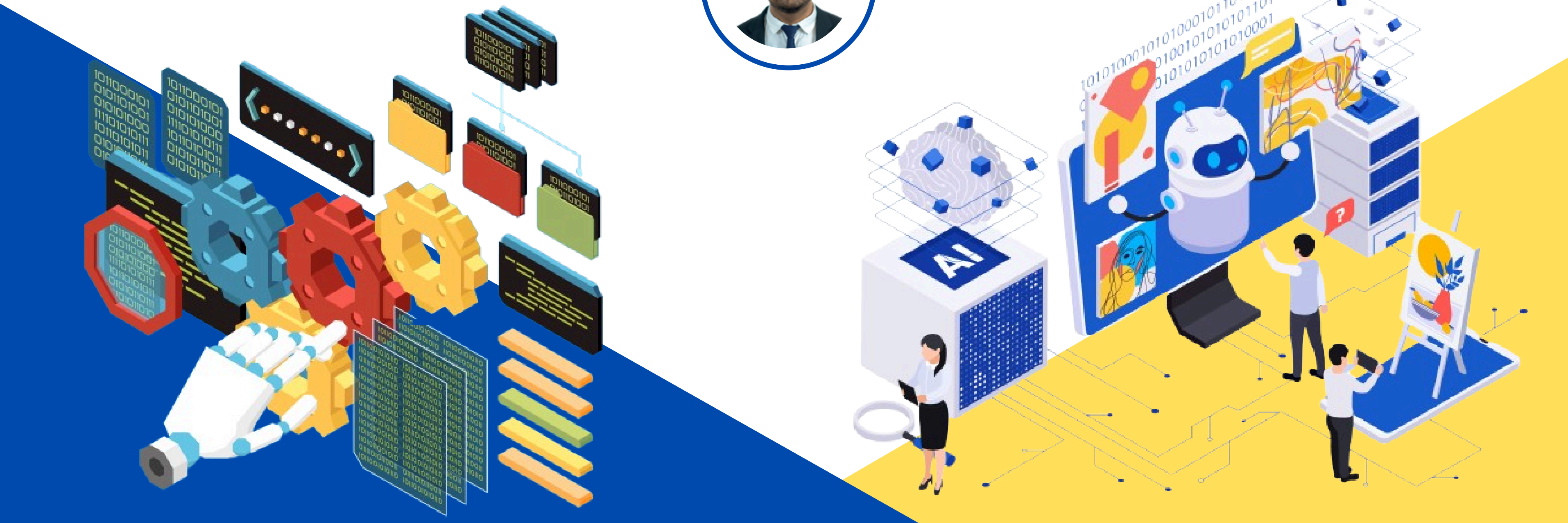
GROUP 10

AI & ML for Business Decisions

March 2026

SDA Bocconi Asia Center, Mumbai | Milan

SRIJAN
3366585



BUSINESS CONTEXT & PROJECT OBJECTIVE

A DATA-DRIVEN APPROACH TO EMPLOYEE RETENTION FOLLOWING M&A

SITUATION

- A healthcare company with 1,220 employees recently completed a merger and acquisition (M&A), incorporating a startup with 250 new employees
- Employee attrition is a critical HR challenge — replacing a single employee costs an average of \$80,000 in recruitment and retraining
- Post-M&A periods are high-risk for attrition — cultural uncertainty and role ambiguity drive voluntary departures
- HR managers need to act proactively — not reactively — to retain top talent

THE OBJECTIVE

- Identify newly acquired employees at high risk of leaving using Machine Learning models
- Classify 250 new employees into attrition risk categories: Low / Medium / High
- Prioritise retention interventions for employees with the highest potential and highest risk
- Translate model predictions into a concrete, data-driven HR retention strategy

Cost of False Negative (missed leaver): **\$80,000** |
Cost of False Positive (unnecessary intervention): **\$12,000** |
Ratio: FN is 6.7× more costly → model must prioritise sensitivity

DATASET OVERVIEW

	Variable	type
1	Attrition	Target(yes/no
2	Business Travel	Categorical
3	Department	Categorical
4	Distance From Home	Numeric
5	Education	Ordinal(1-5)
6	Education Field	Categorical
7	Environment Satisfaction	Ordinal(1-5)
8	Gender	Categorical
9	Job Involvement	Ordinal(1-5)
10	Job Level	Ordinal(1-5)
11	Job Role	Categorical
12	Job Satisfaction	Ordinal
13	Marital status	Categorical
29	Employee Potential	Scoring Dataset Only

DATASET 1 — DEVELOPMENT DATA

- 1,220 rows × 28 variables
- Contains target variable: Attrition (Yes/No)
- Used to train, validate and test ML models
- 70% training / 30% test split (stratified)

DATASET 2 — SCORING DATA

- 250 rows × 28 variables
- No target variable — predictions applied here
- Contains Employee_Potential (High / Low/Medium)
- Used to classify new employees by attrition risk

14	Monthly Income	Numeric
15	NumCompaniesWorked	Numeric
16	Overtime	Categorical
17	% salary hike	Numeric
18	performance Rating	Ordinal
19	Relationship Satisfaction	Ordinal(1-4)
20	Stock Option Level	Ordinal(0-3)
21	Total Working Years	Numeric
22	Training times last year	Numeric
23	Work life balance	Ordinal(1-4)
24	Years At company	Numeric
25	years in current role	Numeric
26	Years Since last promotion	Numeric
27	Years with Current manager	Numeric

- Attrition — Target variable. Present only in development dataset.
- Binary: Yes (left) / No (stayed). 16.1% Yes vs 83.9% No → **Class imbalance noted.**

- Employee_Potential — Present only in scoring dataset.
- Values: **High/Low/Medium** → Used in retention strategy prioritisation.

EXPLORATORY DATA ANALYSIS — TARGET VARIABLE

1,220 EMPLOYEES

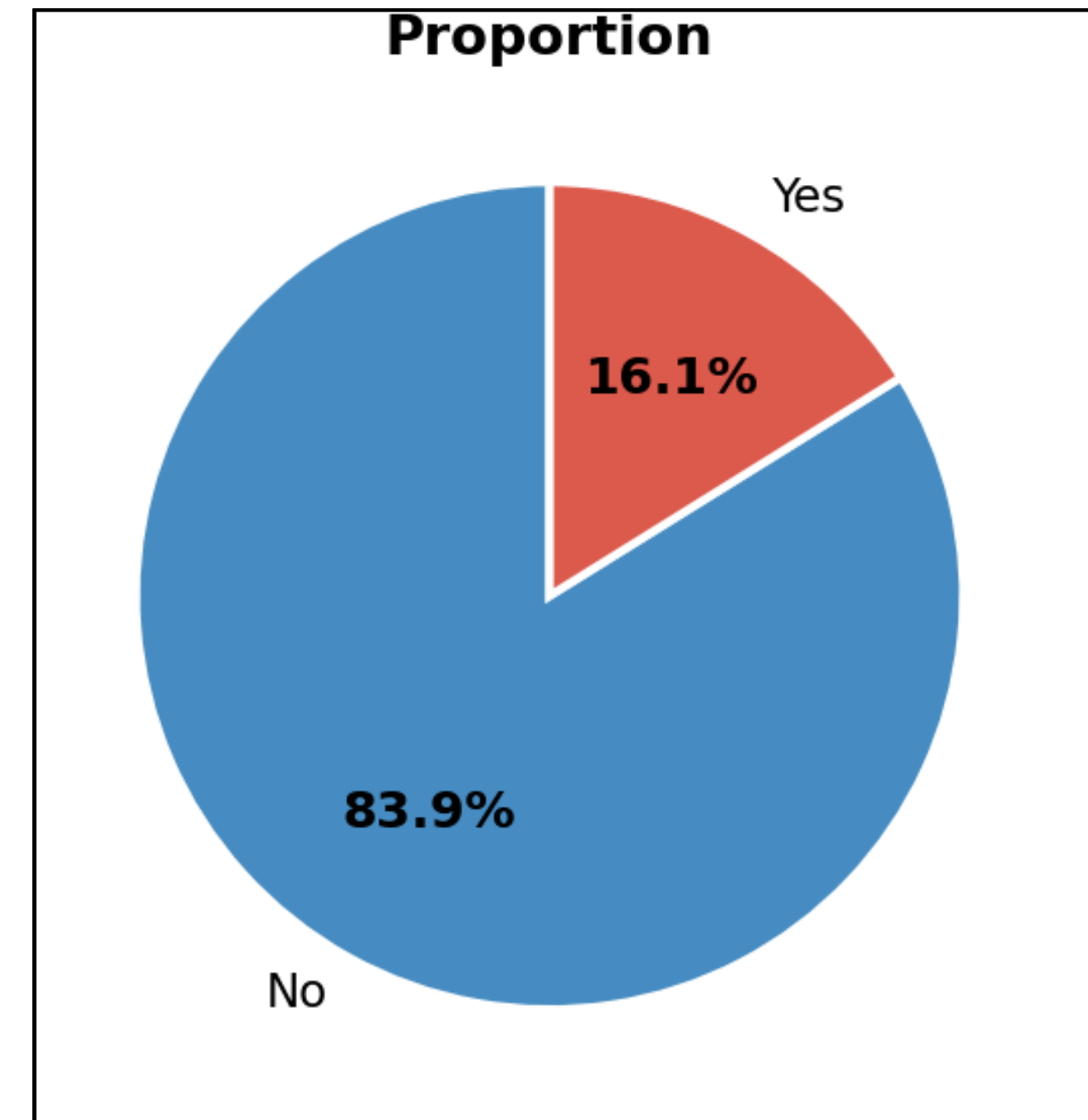
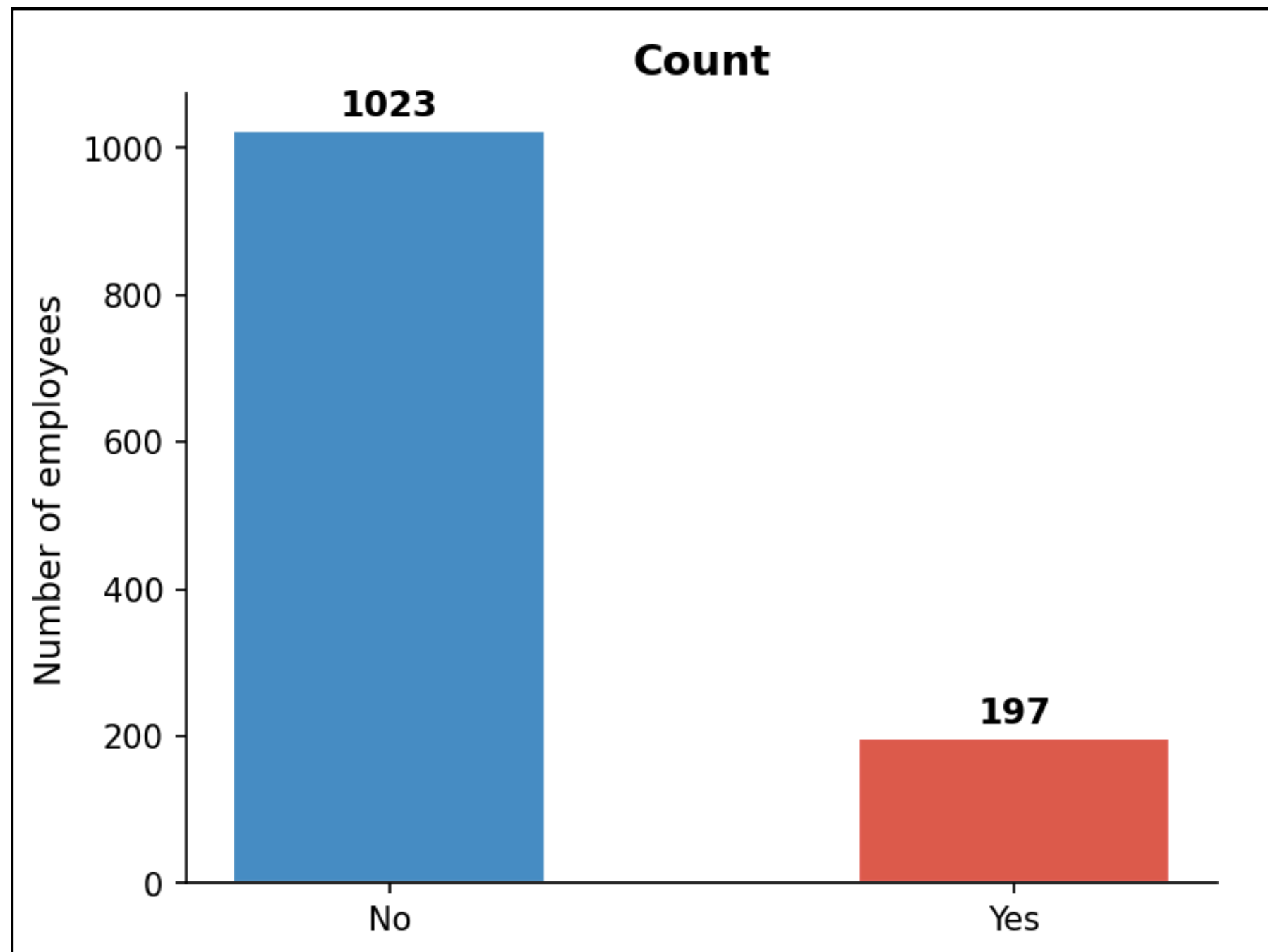
- 1,023 stayed (83.9%)
 - 197 left (16.1%)
- Development dataset total

CLASS IMBALANCE DETECTED

Minority class (Yes) is only 1 in 6 employees — a naive model predicting "No" always scores 83.9% accuracy yet identifies zero leavers

IMBALANCE HANDLED

Oversampling **2x** applied during training to ensure the model learns from attrition cases adequately



EDA — CATEGORICAL VARIABLES VS ATTRITION

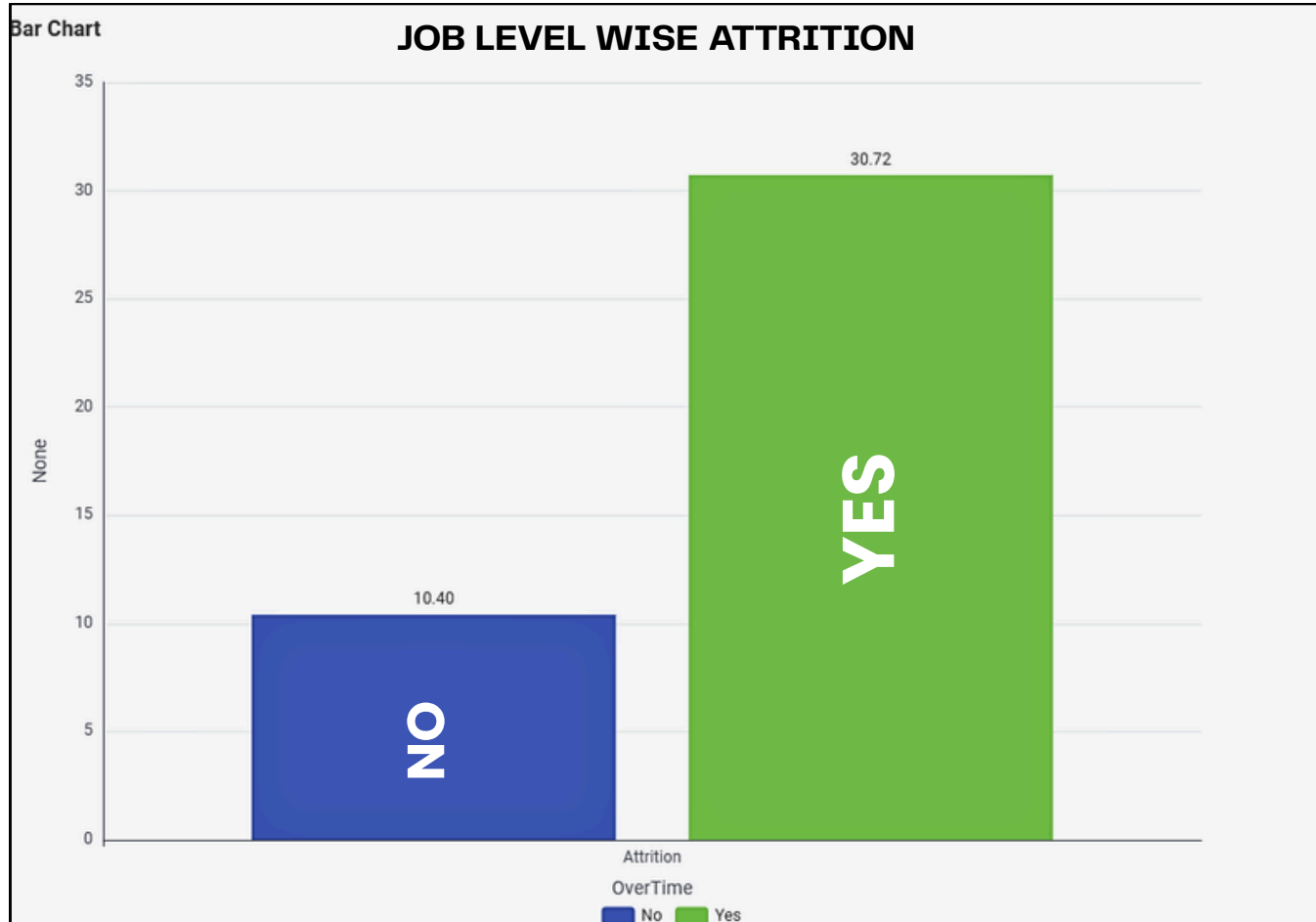
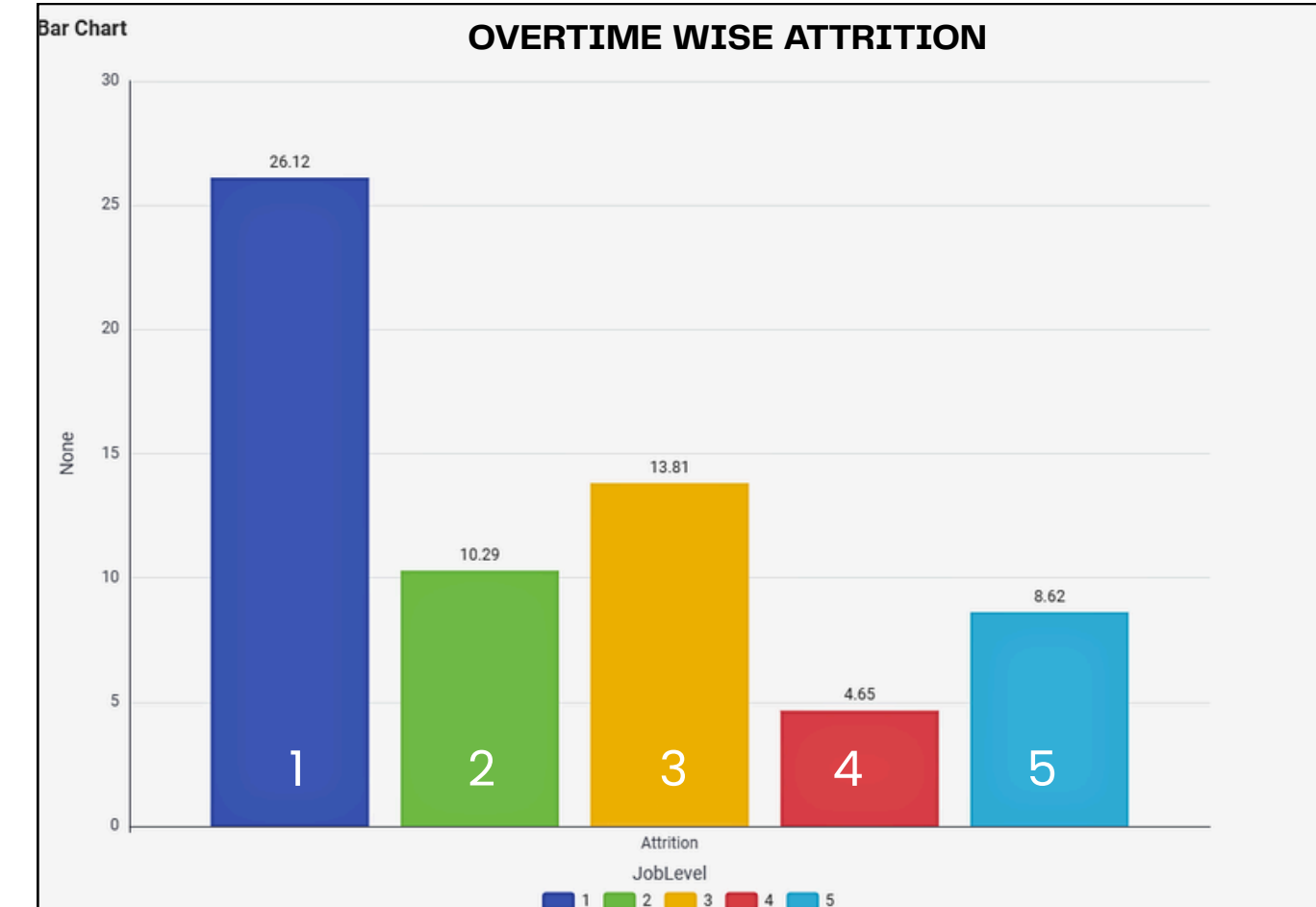
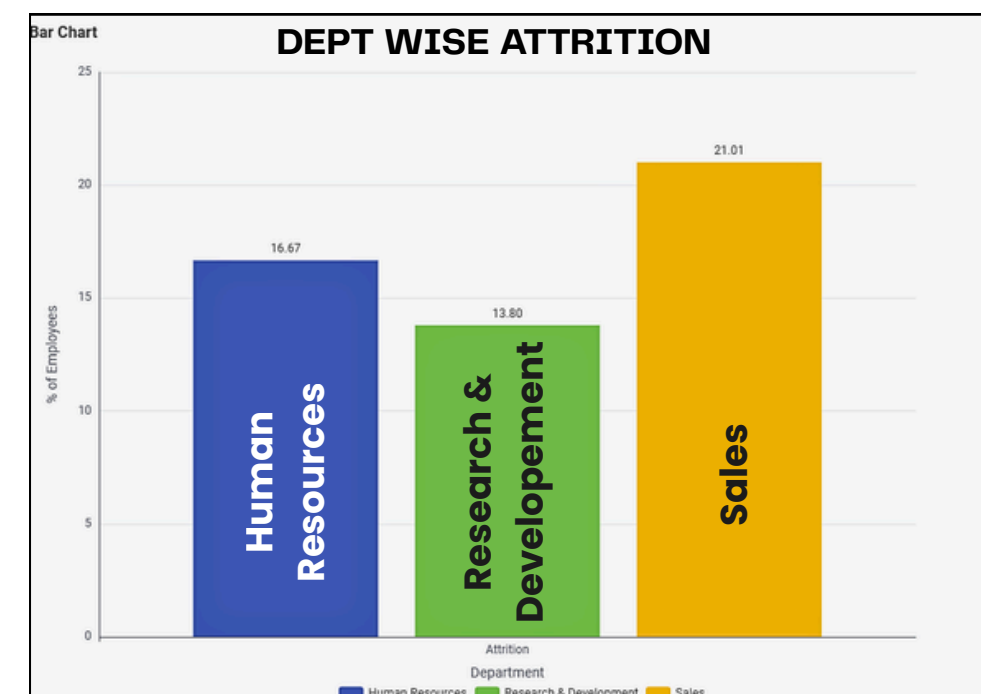
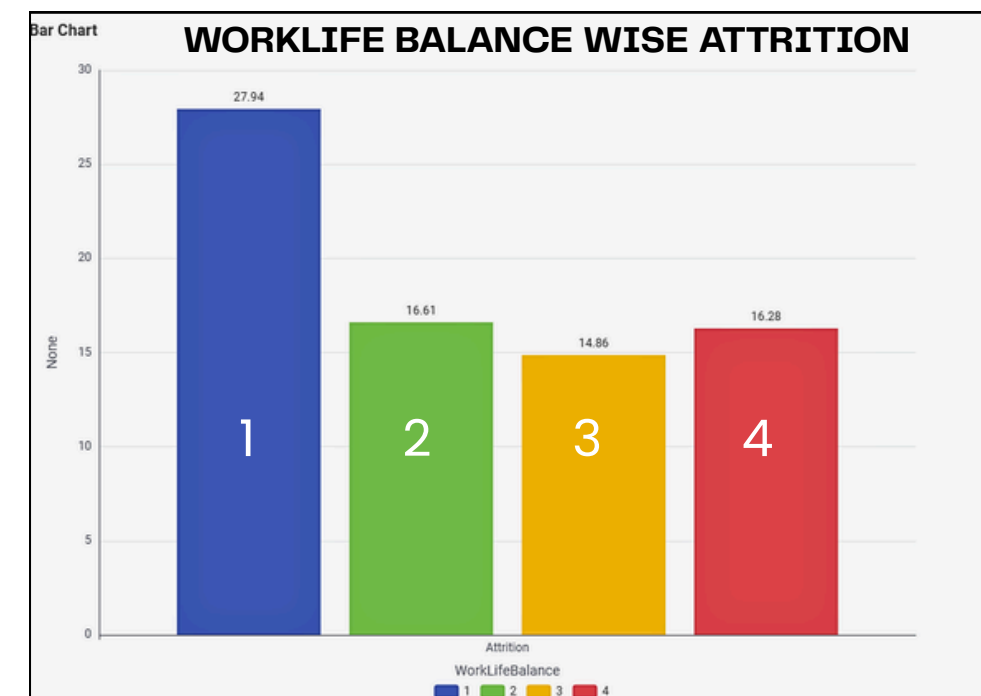
Bivariate analysis — attrition rate (%) within each category group

KEY FINDINGS

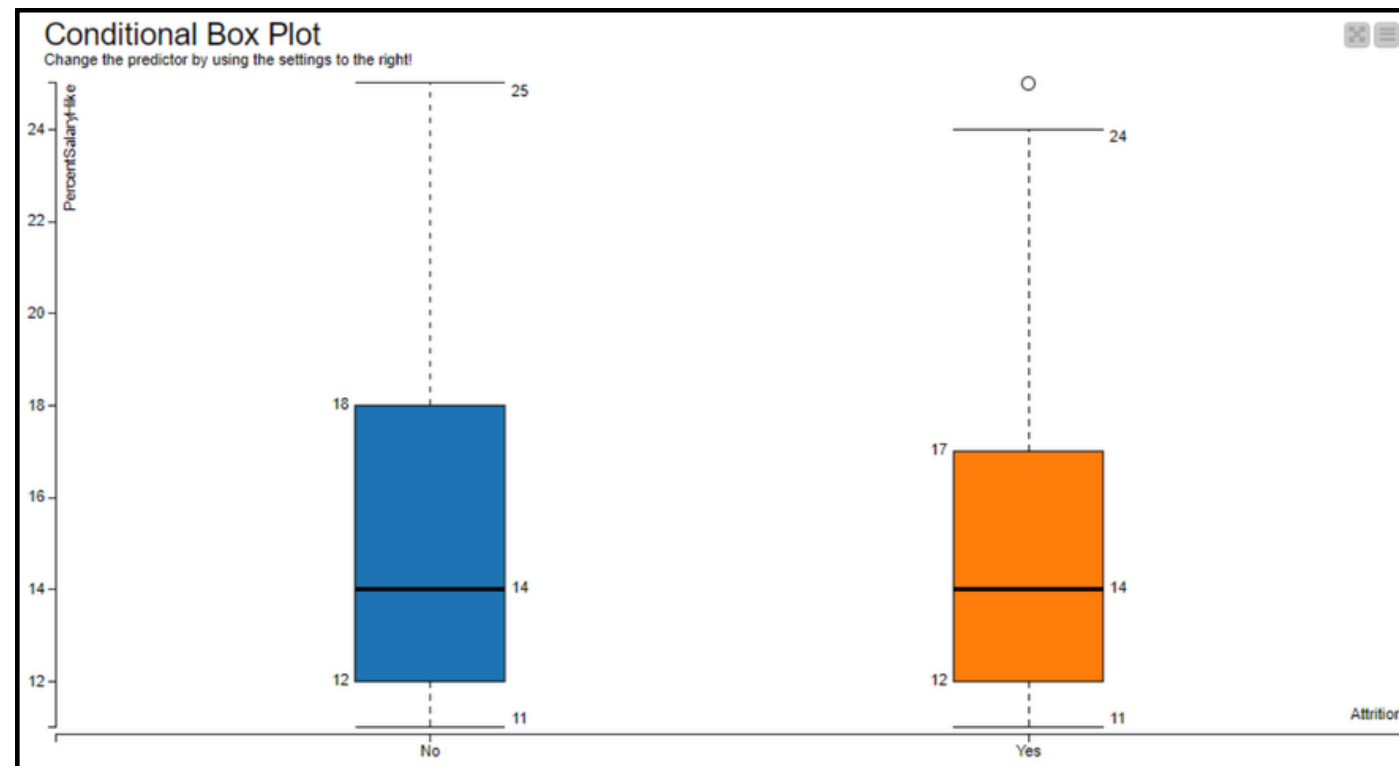
- OverTime is the strongest categorical signal Employees working overtime leave at 30.7% vs only 10.4% for those who don't — nearly 3× higher
- Job Level 1 (entry-level) has the highest attrition rate at 26.1% — senior employees (levels 4–5) are far more stable at under 10%
- Sales department shows the highest attrition at 21.6% vs R&D at 13.8% and HR at 16.7%

BUSINESS INTERPRETATION

- Overtime is a controllable variable — HR can directly reduce attrition risk by managing workload distribution
- Entry-level employees need targeted onboarding and early career development programs to improve retention
- Sales roles carry inherent attrition risk due to performance pressure and market competition — requires role-specific retention strategies



EDA — NUMERIC VARIABLES VS ATTRITION

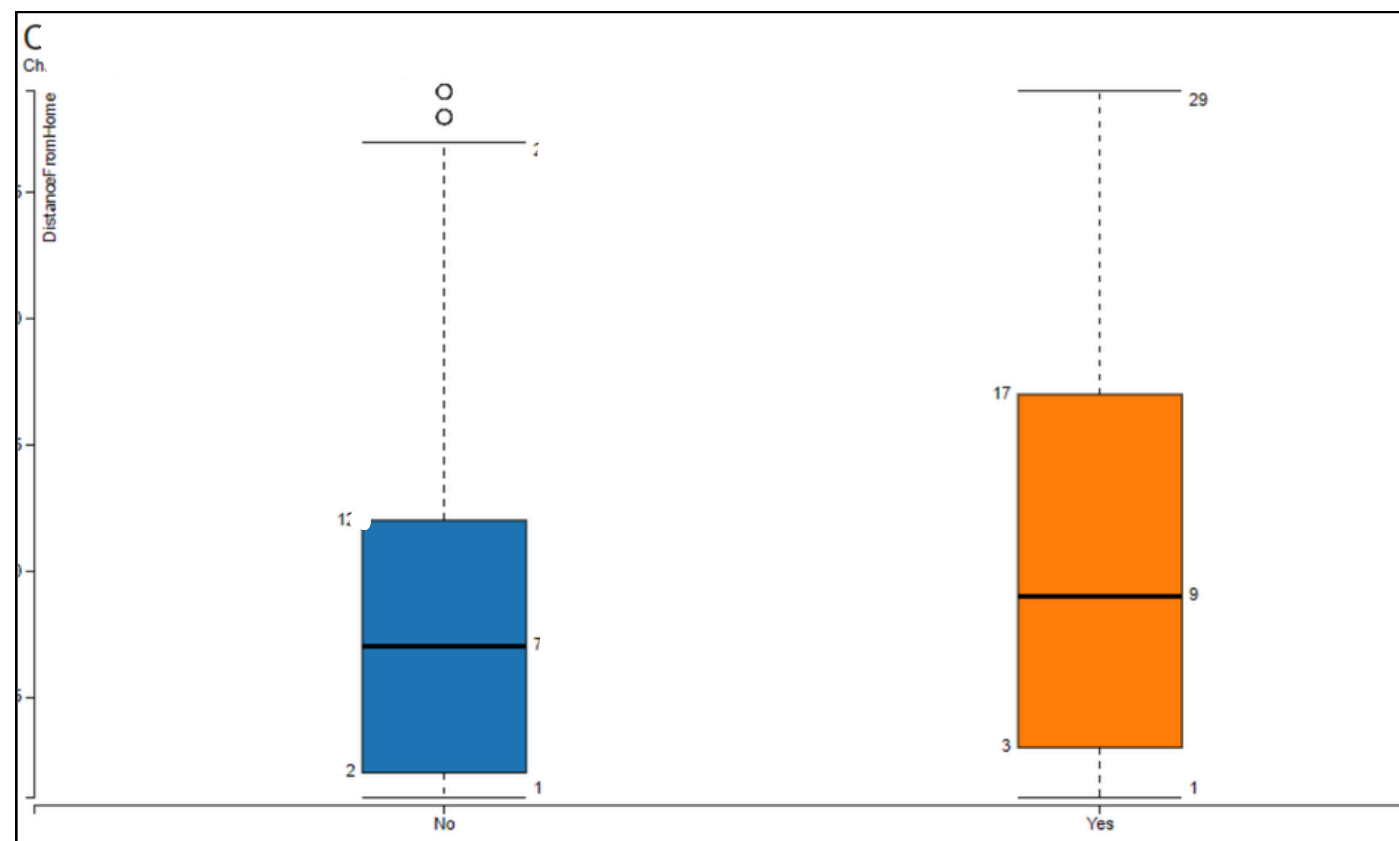
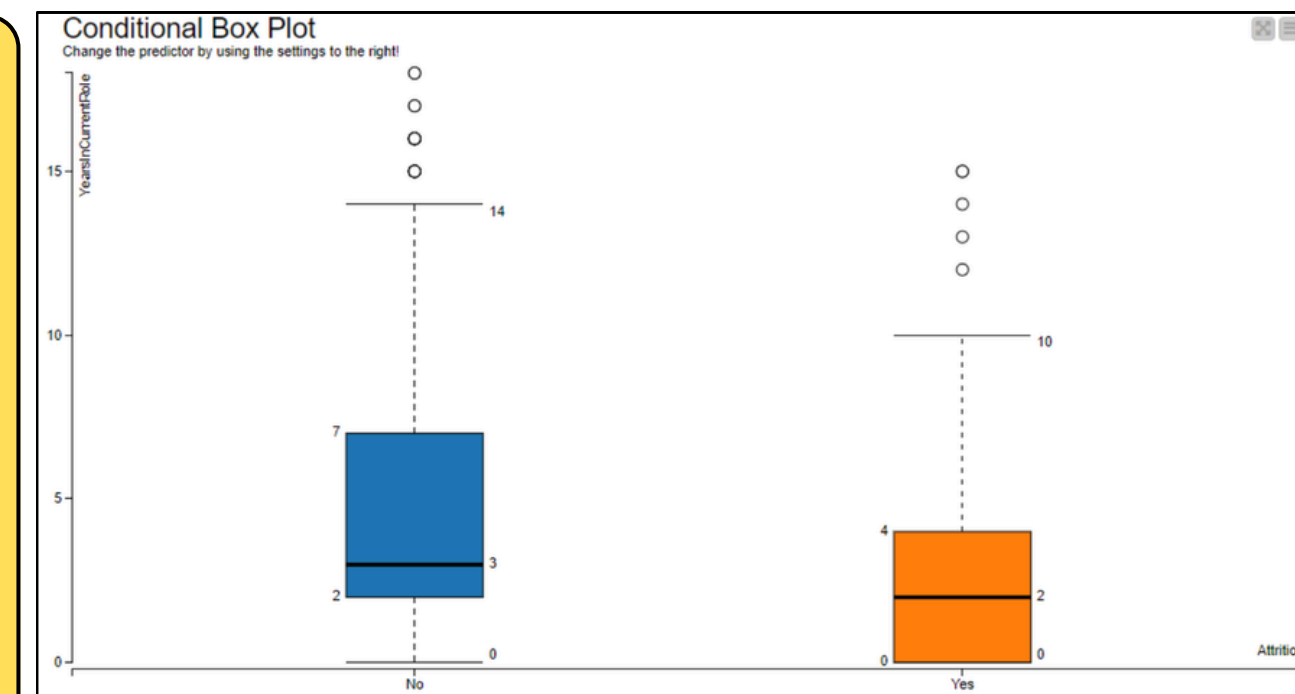


PercentSalaryHike — raise signal

- Stayers median: 14% hike
- Leavers median: 14% hike
- No difference between groups
- Raise % does NOT predict attrition — it is the absolute income level that matters

YearsInCurrentRole — tenure signal

- Leavers median: 2 years in role
- Stayers median: 3 years in role
- Shorter role tenure = higher flight risk

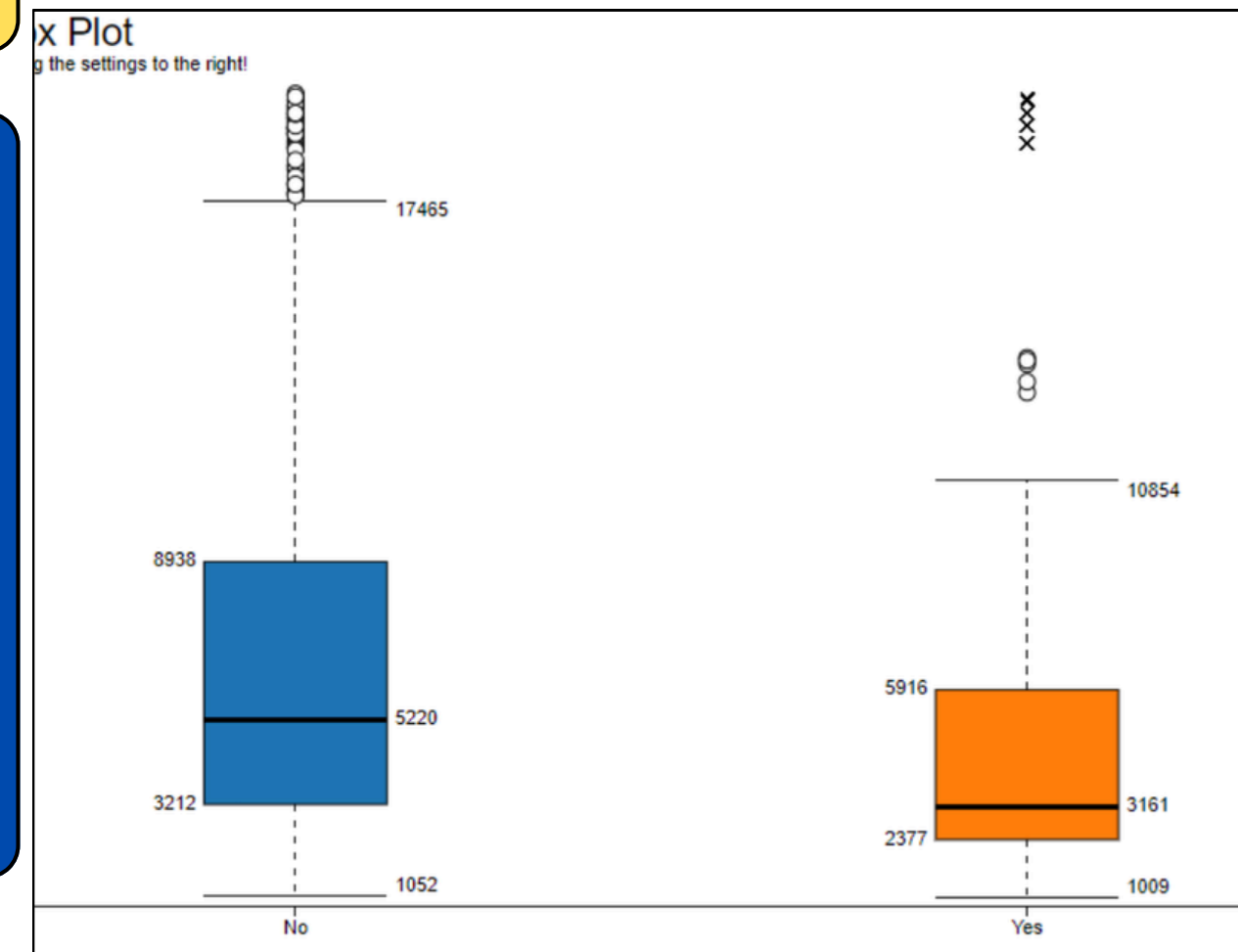


Distance From Home — commute signal

- Leavers median: 9 km from home
- Stayers median: 7 km from home
- Longer commute correlates with attrition

MonthlyIncome — strongest numeric signal

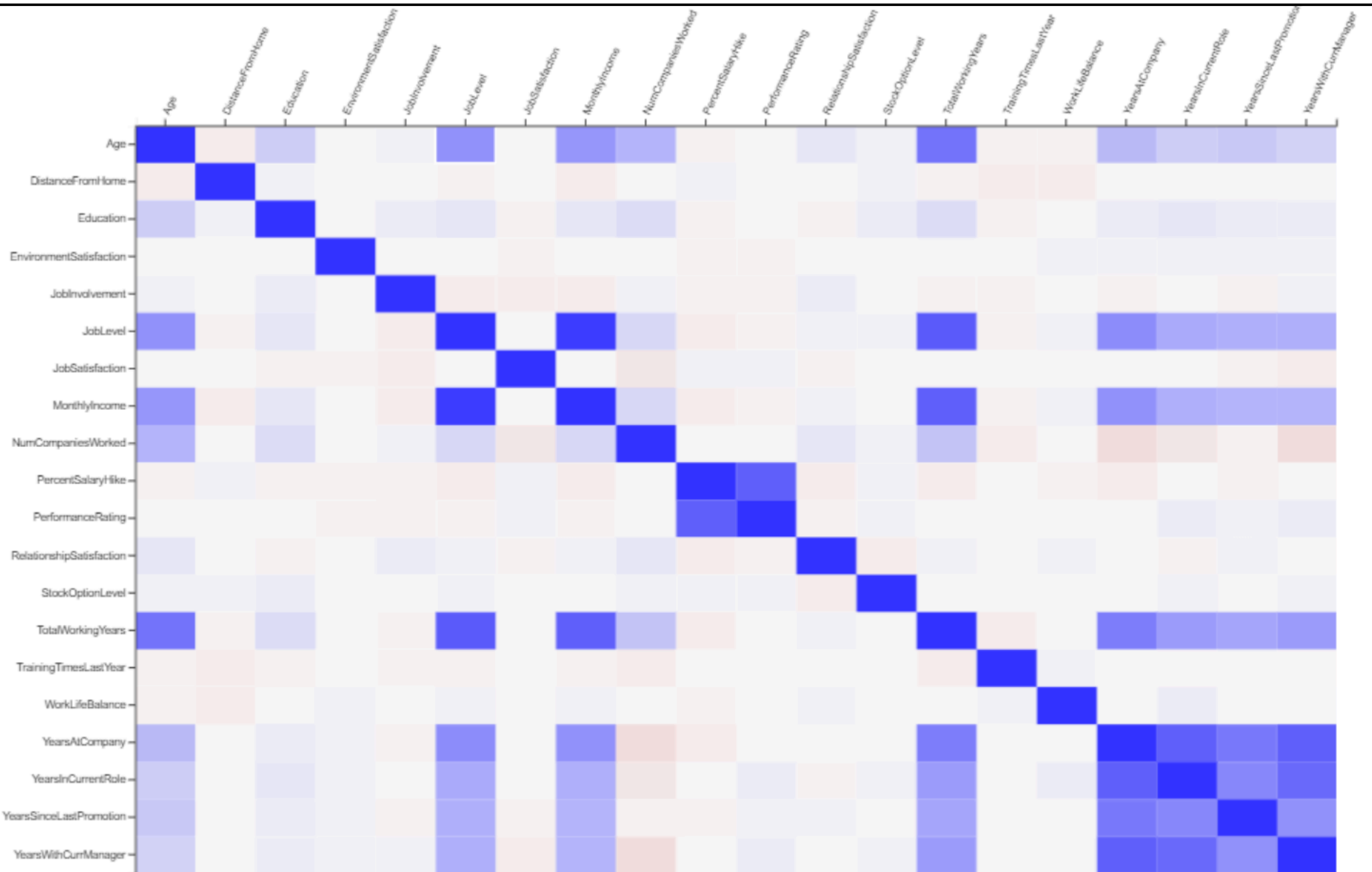
- Median income of leavers: \$3,161
- Median income of stayers: \$5,220
- Leavers earn 39% less than stayers



KEY TAKEAWAY

- Income is the clearest lever — employees earning below \$3,500/month are at significantly elevated risk regardless of other factors
- Employees stuck in the same role for under 2 years need early engagement — either a development path or a role transition conversation
- PercentSalaryHike shows NO difference (median = 14% for both groups) — it is the absolute income level that matters, not the raise percentage

CORRELATION HEATMAP AND EDA SUMMARY



Tenure variables are heavily intercorrelated — tree models will naturally select the strongest one rather than using all four

MonthlyIncome and JobLevel carry overlapping information — IncomePerYearExp engineered feature helps separate these signals

No strong negative correlations found — no variable actively suppresses another in this dataset



DATA PREPARATION

COLUMN FILTER

Tool: Column Filter node

- **Removed: ID column**
(row identifier — zero predictive value)
- Retained: all 27 remaining variables including ordinal scales and categorical strings
- Tree-based models handle categorical variables natively — no encoding required

FEATURE ENGINEERING

Tool: Math Formula + Rule Engine nodes

TenureRatio

- $\text{YearsAtCompany} / (\text{TotalWorkingYears} + 1)$
Captures loyalty relative to total career

SatisfactionScore

- Average of JobSatisfaction, EnvironmentSatisfaction, WorkLifeBalance, RelationshipSatisfaction — composite wellbeing signal

IncomePerYearExp

- $\text{MonthlyIncome} / (\text{TotalWorkingYears} + 1)$
Flags employees underpaid for their experience

NORMALISATION

Tool: **Normalizer node**

- Method: **Min-Max scaling (range 0 to 1)**
- Applied to all continuous numeric columns:
Age, MonthlyIncome, DistanceFromHome, TotalWorkingYears, YearsAtCompany, TenureRatio, SatisfactionScore, IncomePerYearExp and others
- Categorical columns excluded — string variables not normalised

CLASS IMBALANCE HANDLING — OVERSAMPLING 2X (OPTION 1)

PROBLEM

- Training data: 1,023 No vs 197 Yes (83.9% / 16.1%)
- Without rebalancing, model learns to predict "No" for almost everyone and misses leavers entirely

SOLUTION

- EqualSizeOversampling node applied to training set only
- Minority class (Yes) duplicated 2x → ~394 Yes cases
- Test set kept untouched at original distribution
- Option 3x rejected — risks overfitting on 1,220 rows

WHY NOT 3X

- 1,220 rows is a relatively small dataset
- 3x oversampling creates too many duplicates
- Model memorises minority cases rather than generalising — inflated training scores, poor test scores

MODEL TRAINING SETUP

THREE MODELS TRAINED ON REBALANCED DATA WITH A 70/30 STRATIFIED SPLIT

DECISION TREE

Algorithm: Decision Tree Learner

Hyperparameters:

- Min node size: 5 (default 2 — increased to reduce overfitting)
- Quality measure: Gini index
- Pruning method: MDL
- Max depth: unrestricted

Purpose in comparison: Simplest model — provides interpretable baseline. Expected to underperform ensemble methods.

RANDOM FOREST

Algorithm: Random Forest Learner

Hyperparameters:

- Number of trees: 200 (default 100 — increased for stability)
- Features per split: Square root of total features (standard RF setting)
- Sampling: with replacement
- Voting: majority class

Purpose in comparison: Ensemble of decorrelated trees. High accuracy expected. Strong specificity profile

GRADIENT BOOSTED TREES

Algorithm: GBT Learner

Hyperparameters:

- Max tree depth: 5 (range 3–7, default kept)
- Learning rate: 0.05 (reduced from 0.1 for better generalisation)
- Iterations: 300 (default 100 — increased to complement lower LR)
- Subsample: 0.8

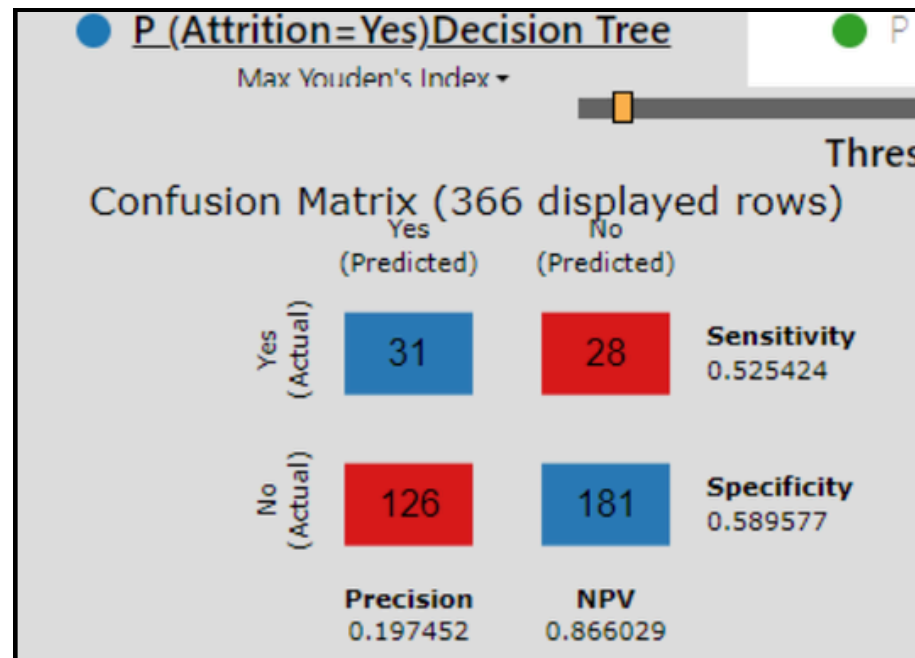
Purpose in comparison: Sequential boosting model. Best at capturing complex patterns. Prioritises sensitivity over accuracy.

Model	Expected strength	Expected weakness
Decision Tree	Interpretability	Overfitting, low AUC
Random Forest	Accuracy, stability	Lower sensitivity
GBT	Sensitivity, AUC	Lower overall accuracy

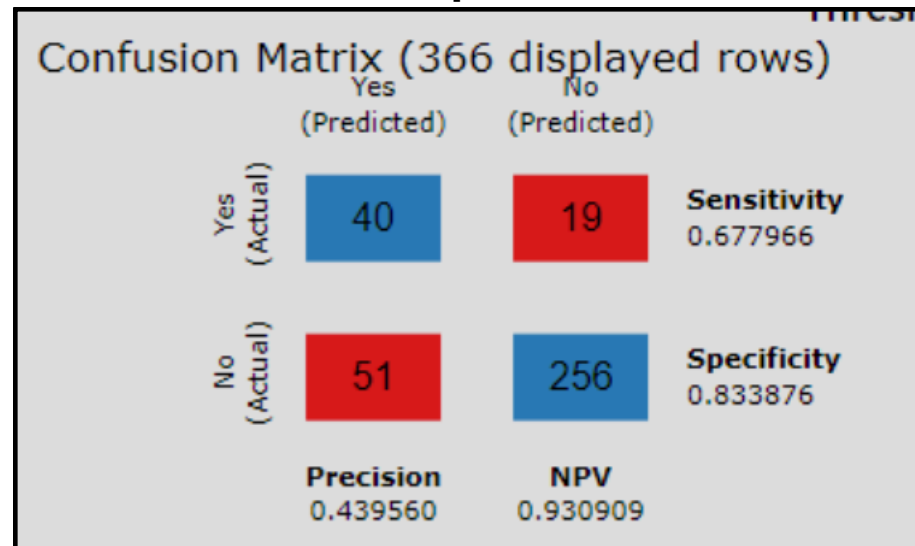
MODEL EVALUATION

366 test rows | Threshold set to maximise Youden's Index per model

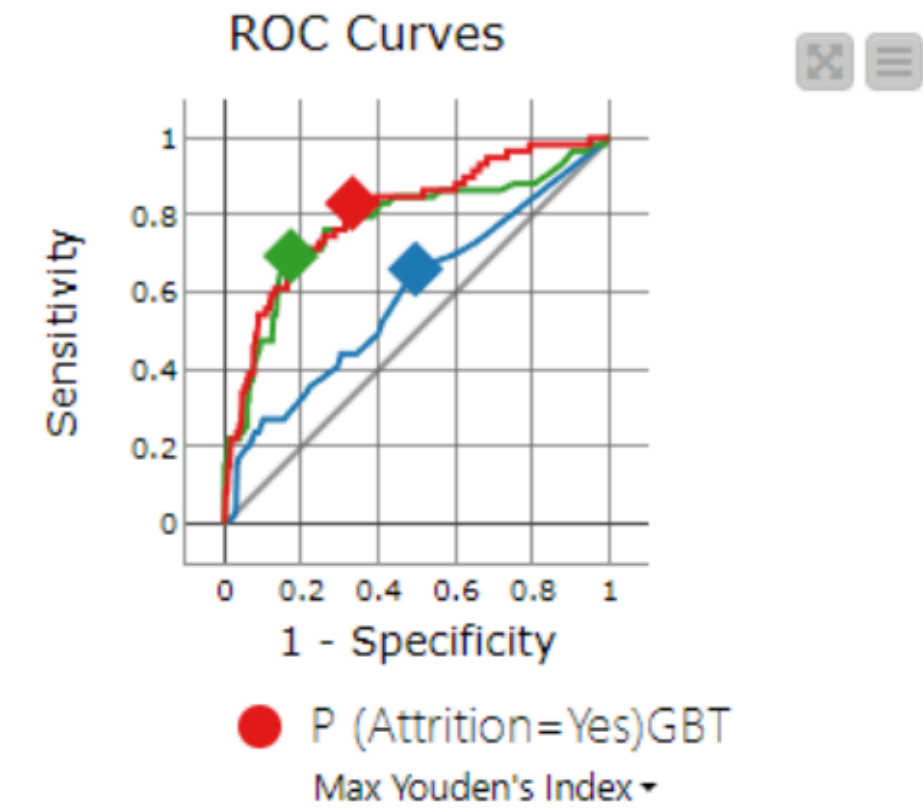
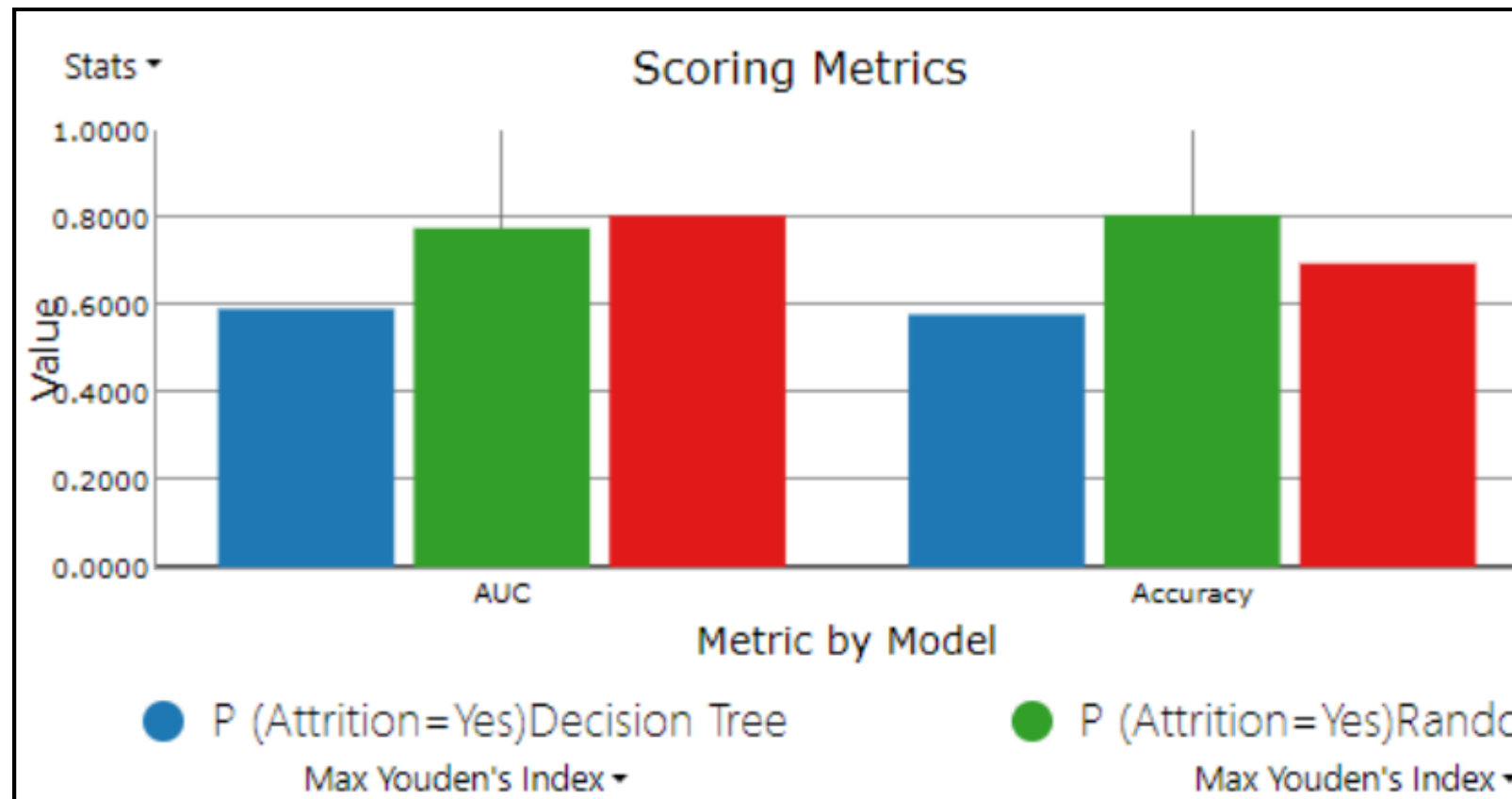
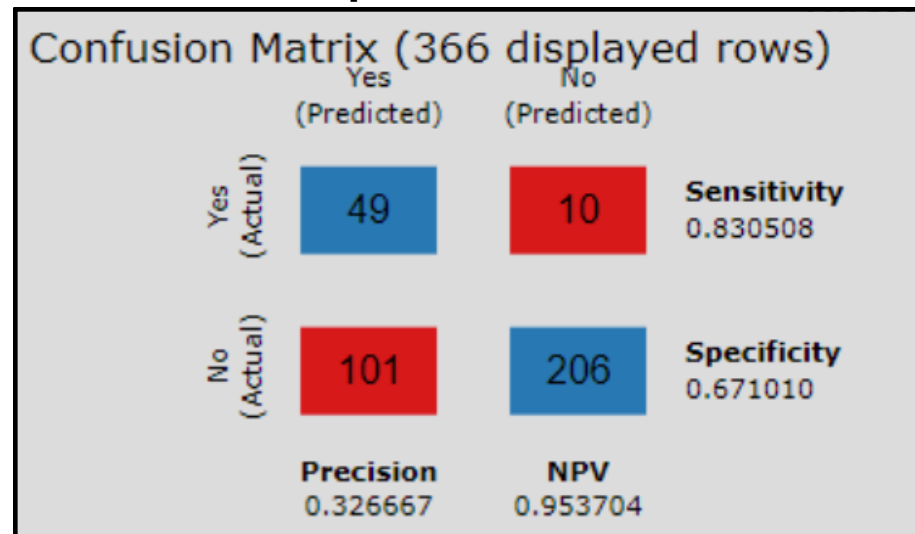
Decision Tree | Threshold: 0.033



Random Forest | Threshold: 0.260



GBT | Threshold: 0.012



- Blue = True Positive / True Negative (correct predictions)
- Red = False Positive / False Negative (errors)
- FP cost: \$12,000 per employee | FN cost: \$80,000 per employee

Why different thresholds per model?

- Default threshold = 0.5 is designed for balanced classes
- Our data is 83.9% No / 16.1% Yes — threshold must be lowered
- Youden's Index finds the optimal threshold that maximises Sensitivity + Specificity simultaneously
- DT: 0.033 | RF: 0.260 | GBT: 0.012
- Lower threshold = more employees flagged as at-risk = higher sensitivity = fewer missed leavers

Models performance

Model Name	Default Threshold	Accuracy	Sensitivity	Specificity	AUC	Total Cost
Decision Tree	0.033	0.579	0.525	0.59	0.591	3,752,000
Random_Forest	0.26	0.809	0.678	0.834	0.776	2,132,000
GBT	0.012	0.697	0.831	0.671	0.805	2,012,000

MODEL SELECTION — WHY WE CHOSE GRADIENT BOOSTED TREES

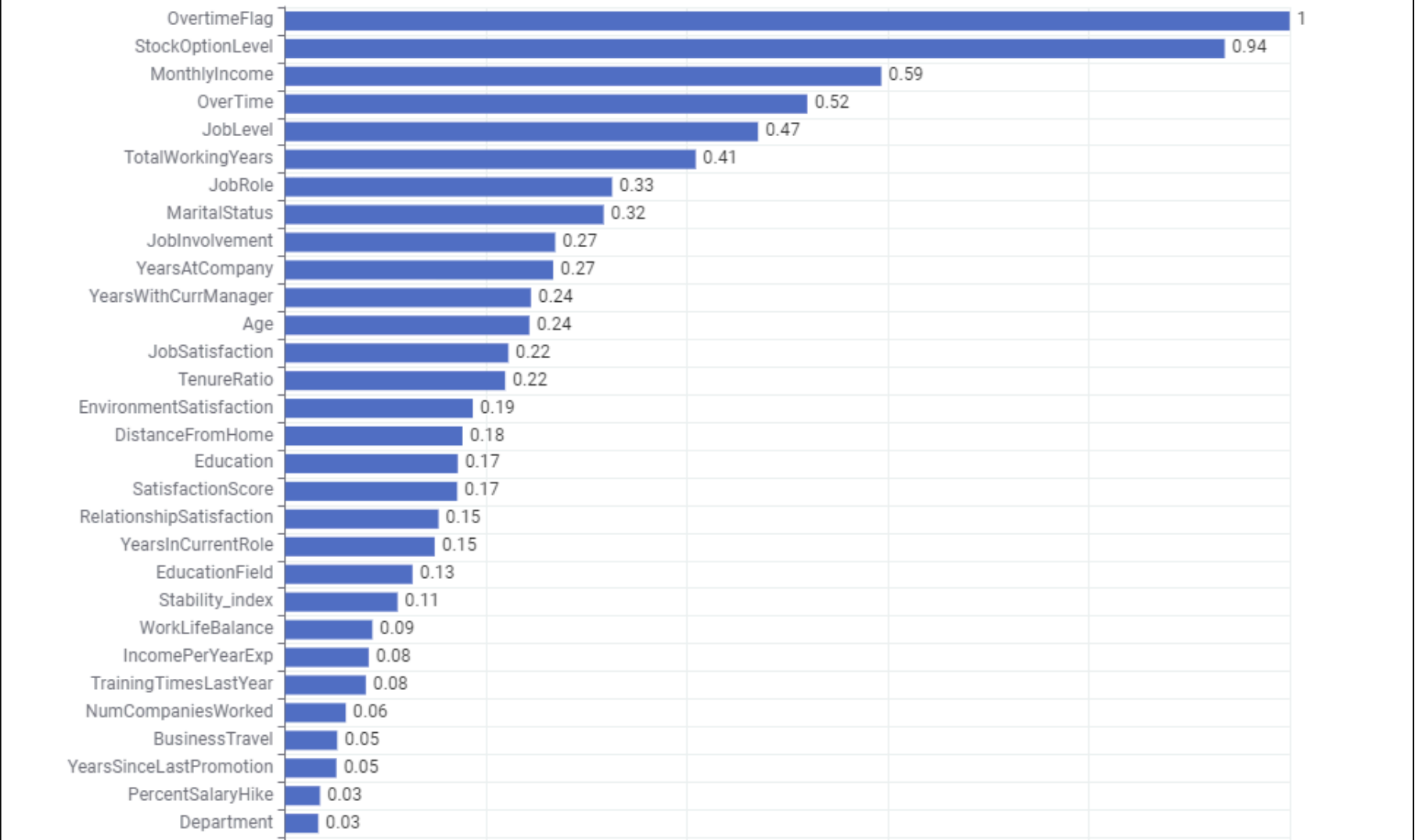
Selection based on AUC, sensitivity and total misclassification cost

DECISION TREE ELIMINATED IMMEDIATELY

- **AUC 0.591** — near random guessing
- **Accuracy 57.9%** — below naive baseline
- **Total cost \$3,752,000** — 86% higher than GBT | No further consideration

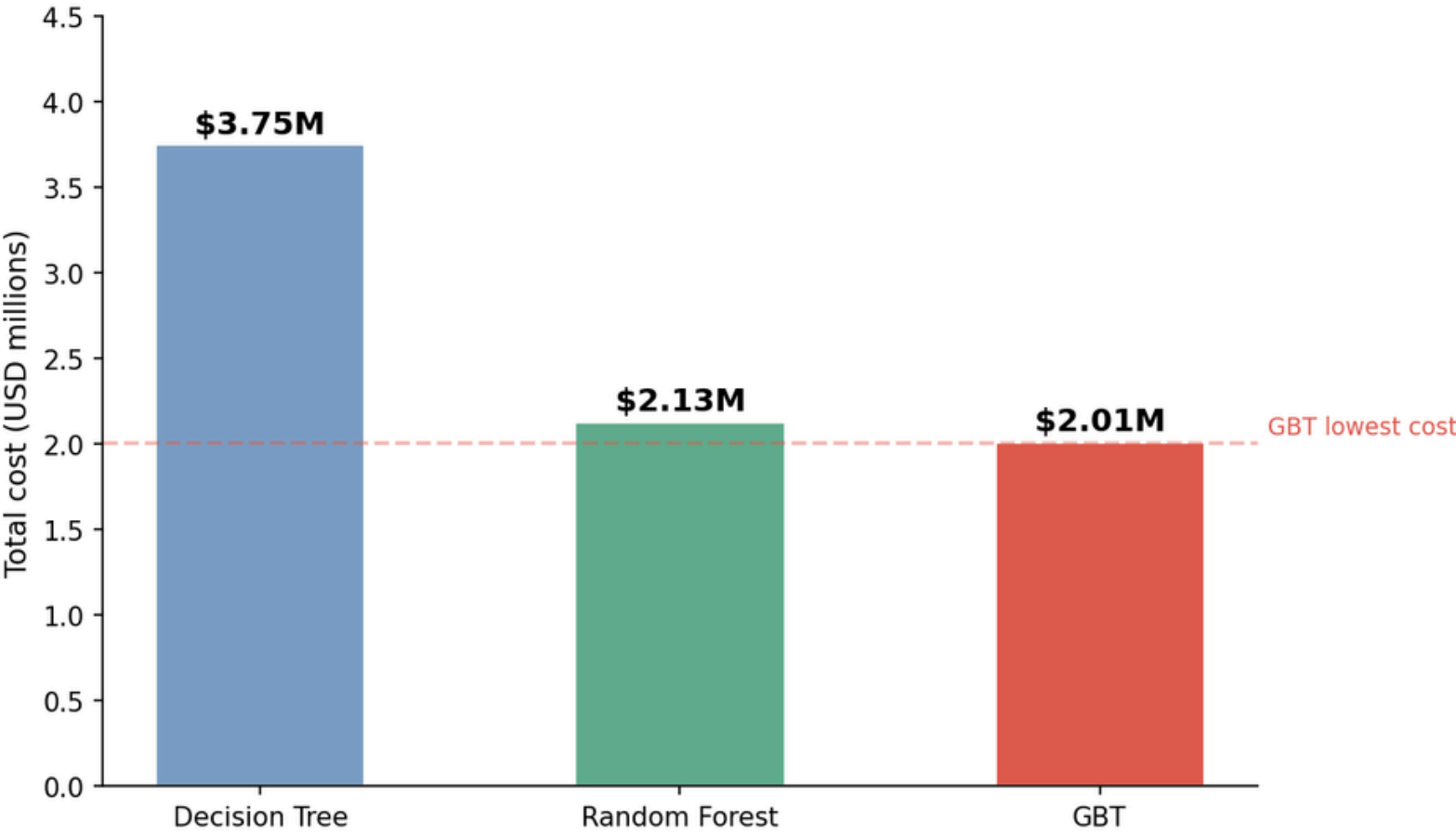
Criterion	Random Forest	GBT	Winner
AUC	0.776	0.805	GBT
Accuracy	80.90%	69.70%	RF
Sensitivity	67.80%	83.10%	GBT
Specificity	83.40%	67.10%	RF
Total cost	\$2,132,000	\$2,012,000	GBT
Leavers correctly caught	46 out of 59	49 out of 59	GBT
False negatives	13 missed leavers	10 missed leavers	GBT
False positives	51 wrong flags	101 wrong flags	RF

RANDOM FOREST Normalized Variable Importance



Accuracy favours RF but is misleading with imbalanced classes — a model predicting No always scores 83.9% accuracy yet catches zero leavers. Sensitivity and cost are the correct business metrics here.

Total Misclassification Cost by Model



MODEL INFERENCE & SCORING THE 250 NEW EMPLOYEES

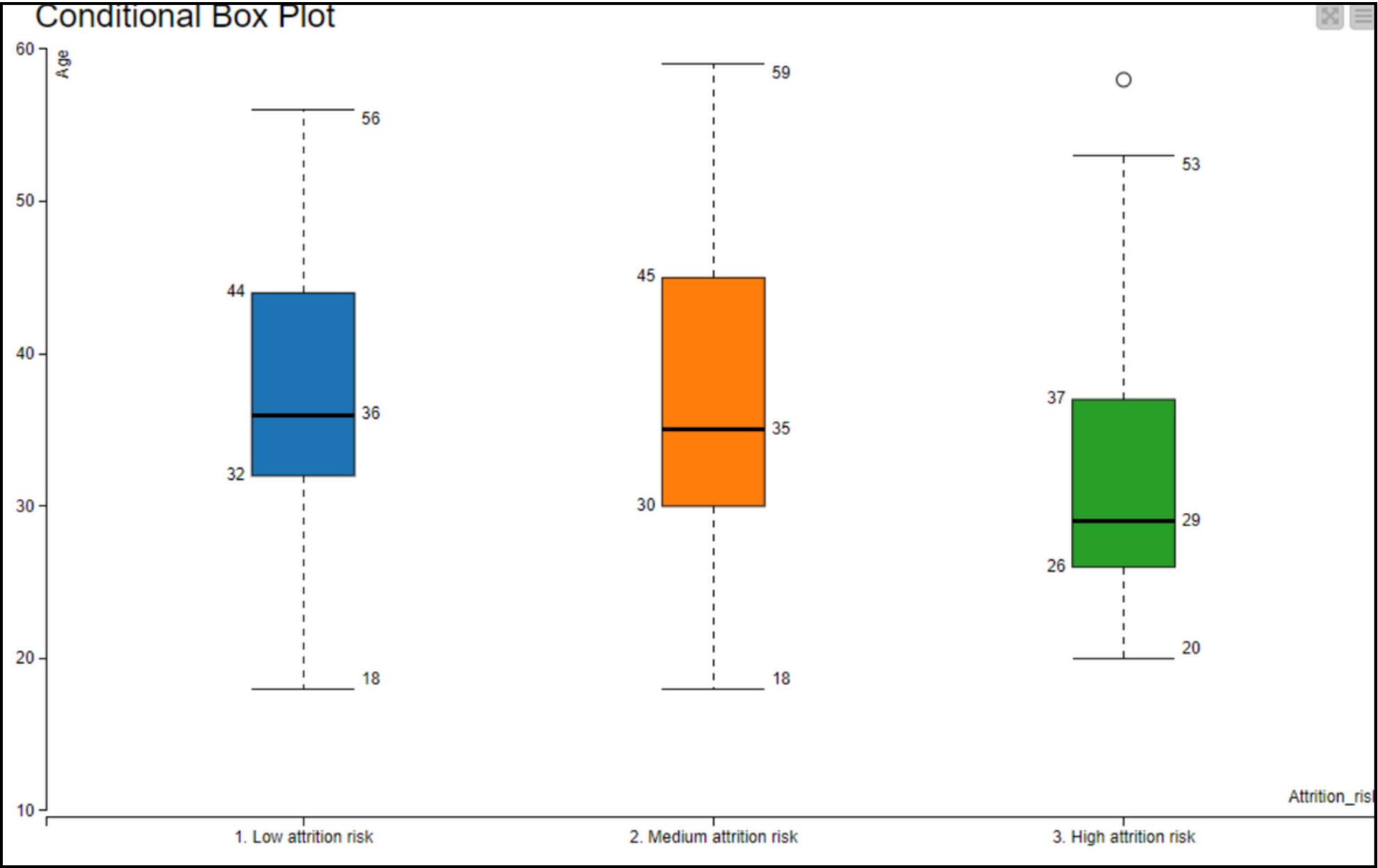
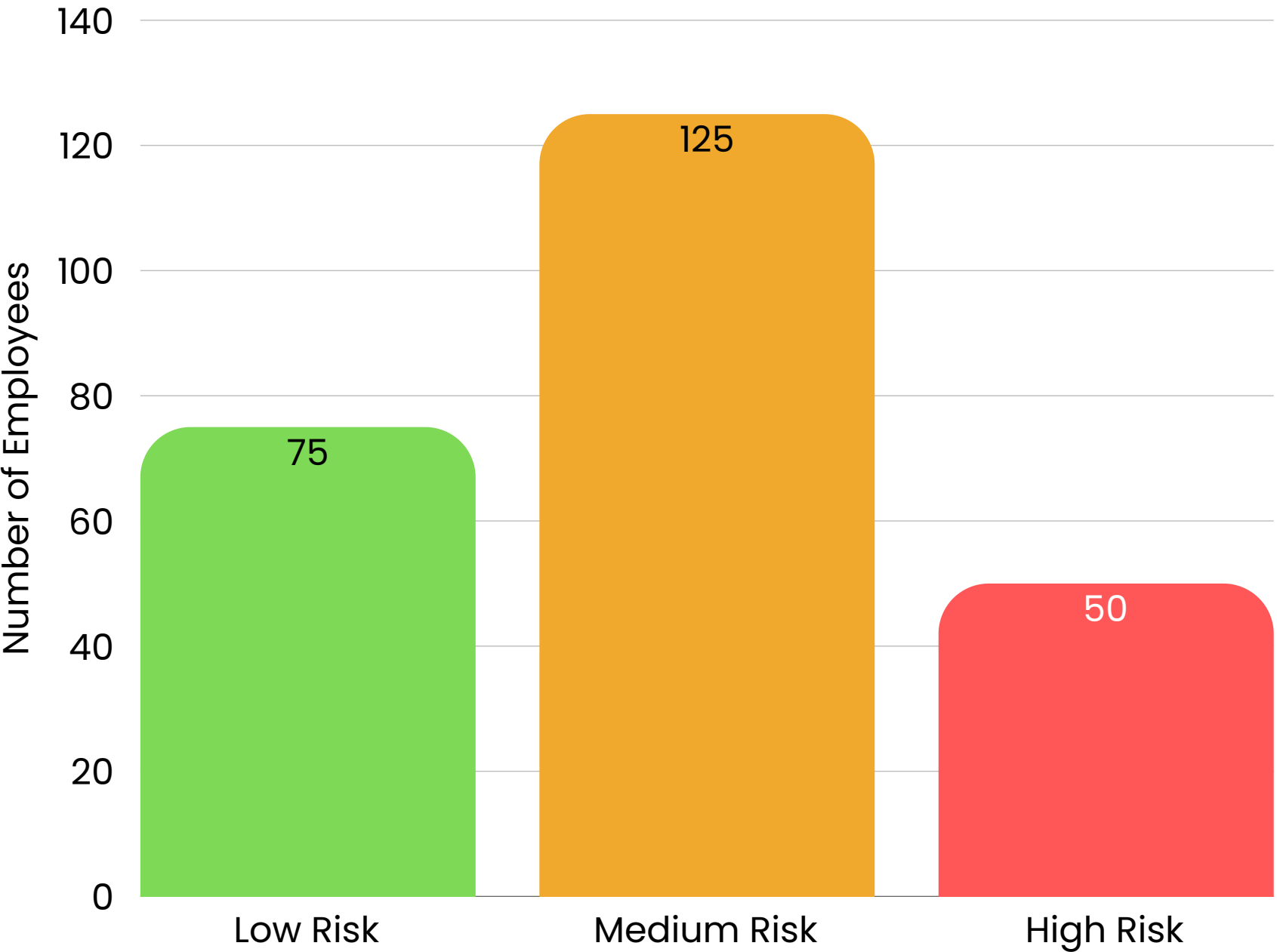
GBT applied to scoring dataset | Employees classified into three attrition risk categories

Cross Tabulation of Attrition_risk by Employee_Potential			
Frequency	High potential	Low/Medium Potential	Total
1. Low attrition risk	33	92	125
2. Medium attrition risk	22	53	75
3. High attrition risk	21	29	50
Total	76	174	250

High Risk + High Potential (21)
Priority retention targets – high risk of leaving AND high future value to company

Medium Risk + High Potential (22)
Prevention segment – intervene now before they drift to high risk

High Risk + Low/Med (29)
Assess role criticality before investing in retention



High risk employees are significantly younger (median age 29) vs low risk (median 36) – youth and mobility drive attrition risk

RETENTION STRATEGY – RISK X EMPLOYEE POTENTIAL MATRIX

Targeted HR interventions based on attrition risk level and employee future value

RISK X POTENTIAL	LOW/MEDIUM POTENTIAL	HIGH POTENTIAL
HIGH RISK	MANAGE CAREFULLY 29 EMPLOYEES - Asses if role is business critical - small retention bonus if hard to replace - Beginn succession planning if non-critical	TOP PRIORITY – ACT NOW 21 EMPLOYEES AVG AGE – 29 59% ON OVERTIME - Immediate salary benchmarking + adjustment - Remove or compensate excessive overtime - Personal career roadmap + senior mentor
MEDIUM RISK	STANDARD ENGAGEMENT 53 EMPLOYEES - Quarterly satisfaction pulse surveys - Team culture and manager check-ins - Monitor – flag if risk escalates	DEVELOP AND ENGAGE 22 EMPLOYEES AVG TENURE 4.7 YEARS - Stretch assignments and new projects - Formal mentoring + promotion criteria shared - Leadership developement track enrollment
LOW RISK	MONITOR ONLY 92 EMPLOYEES - No urgent action required - Annual engagement survey suffic	LONG TERM INVESTMENT 33 EMPLOYEES - Plan long term growth and career paths - Keep engages with challenging work - Pipeline for future leadership roles

Cost logic: FN = 80, 000(missed leaver) vs FP = 12,000 (unnecessary intervention) | Every 1 spent on targeted retention saves upto 5.30 in replacement costs

RETENTION ROI – CASE FOR PROACTIVE HR INVESTMENT

Targeted HR interventions based on attrition risk level and employee future value

SCENARIO 1 DO NOTHING

~31 HIGH RISK EMPLOYEES LEAVE

\$80,000

PER EMPLOYEE REPLACEMENT

\$2,480,000

TOTAL REPLACEMENT COST

RECRUITMENT + TRAINING + PRODUCTIVITY LOSS
FOR EVERY EMPLOYEE WHO LEAVES UNDETECTED

SCENARIO 2

RETAIN ALL HIGH RISK

31 EMPLOYEES – BLANKET RETENTION

\$15,000

PER EMPLOYEE AVG PACKAGE

\$465,000

TOTAL RETENTION SPEND

SALARY ADJUSTMENT + BONUS + DEVELOPMENT
BUDGET APPLIED TO ALL 31 HIGH RISK EMPLOYEES

SCENARIO 3

SMART TARGETING

HIGH RISK + HIGH POTENTIAL ONLY

\$15,000

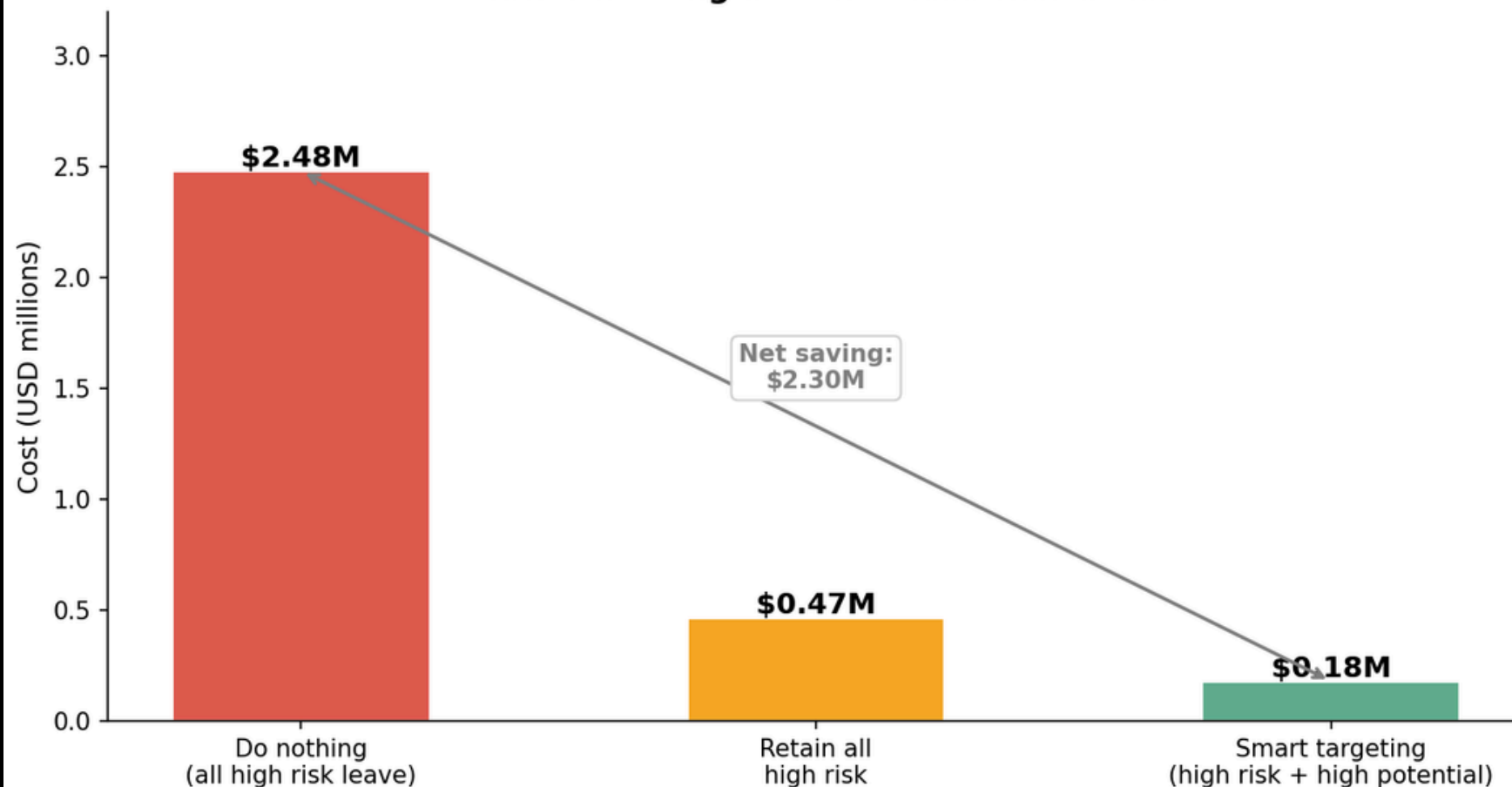
PER EMPLOYEE AVG PACKAGE

\$180,000

TOTAL RETENTION SPEND

SAME RETENTION PACKAGE BUT FOCUSED
ON 12 HIGHEST-VALUE EMPLOYEES ONLY

Retention Programme — Cost Scenarios



5.3X → Return on Investment

\$2,300,000 → Net saving vs do nothing

\$65,000 → Net saving per retained
employee

12 → employees to target

\$15,000 → AVg retention package cost

Business Conclusion:

- The ML model identifies the 12 employees where retention spend is truly justified.
- Spending \$180,000 to retain them avoids \$2,480,000 in replacement costs.
- Without the model, HR would either:
(a) miss leavers entirely – \$2.48M cost
(b) spend broadly – \$465K with low ROI
- Targeted ML-driven retention delivers the highest return at the lowest cost

CONCLUSIONS & LIMITATIONS

Key findings, business implications and honest acknowledgement of model boundaries

MODEL PERFORMANCE

- GBT selected as best model
- AUC: 0.805 – strong discrimination
- Sensitivity: 83.1% – catches 8 in 10 leavers
- Lowest total cost: \$2,012,000
- Outperforms RF and DT on all

KEY ATTRITION DRIVERS

- Overtime: 30.7% vs 10.4% attrition rate
- Monthly income gap: 3,161 vs 5,220
- Entry-level (Job Level 1): 26.1% attrition
- Sales department: 21.6% attrition
- Young + single employees most mobile
- Distance from home adds friction

BUSINESS IMPACT

- 50 high-risk employees identified
- 21 are High Potential – top priority
- Smart retention cost: \$180,000
- Replacement cost avoided: \$2,480,000
- Net saving: \$2,300,000
- ROI of retention programme: 5.3x

MODEL LIMITATIONS

- GBT specificity only 67.1% – 33% of stayers incorrectly flagged as leavers
- GBT threshold 0.012 is very aggressive – flags employees with just 1.2% predicted risk
- Dataset limited to 1,220 employees – larger data would improve generalisation
- No external market data used – salary benchmarks could sharpen income signal

FUTURE WORK

- Integrate external salary benchmarks to sharpen income-based predictions
- Re-train model quarterly as new employee data becomes available
- Add exit interview data as features to capture qualitative leaver signals
- Extend model to predict time-to-attrition benchmarks could sharpen income signal, not just binary Yes/No classification

Thank You